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Algorithm Delegation and Responsibility: Shifting Blame to the Programmer ?

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Abstract

In a lab experiment, we investigate whether individuals delegate allocation decisions to an intermediary in order to shift blame. The intermediary can be a human, a rule-based algorithm (RA), or an artificial intelligence algorithm (AI), depending on the treatment condition. Behind these algorithms, a programmer fully controls the decisions of the RA and partially controls the decisions of the AI. We find that participants delegate regardless of the type of intermediary (human, RA, or AI). While human intermediaries and RA programmers are perceived as more responsible for inequalitarian decisions, the delegator is perceived as less responsible. AI programmers and the delegator are not perceived as responsible for the inequalitarian allocations made by the AI. This allows AI programmers to exploit the moral wiggle room and select inequalitarian allocations more frequently.

JEL Codes: C92, D63

Keywords: Artificial Intelligence, Delegation, Responsibility, Moral Wiggle Room, Punishment, Laboratory Experiment

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1 Introduction

In an era where algorithms shape everything from our social media feeds to the financial markets, understanding our interactions with these digital decision-makers is more important than ever. As algorithms evolve to become our advisors, employees and even our managers, the lines between human judgement and machine calculation are blurring, raising profound questions about trust, responsibility, and ethics in our increasingly automated world. Organizations are being transformed to welcome these algorithms (Shrestha et al., 2019; Langer and Landers, 2021; Agrawal et al., 2024). At the same time, a growing number of studies are discussing the pros and cons of these autonomous algorithms in different tasks and domains such as autonomous vehicles (Bonnefon et al., 2016; Shariff et al., 2017; Awad et al., 2020).¹ What all algorithms have in common is that there is always a principal (also referred to as “*the user*”) behind them who chooses to delegate her decision to the algorithm. This algorithm is linked to “*a programmer*” who manipulates and develops it.

A principal may choose to delegate a decision for several reasons, including the expertise of the agent, the inability to manage everything, to save time, or simply because she does not want to do it herself. Bartling and Fischbacher (2012) also show that a principal can delegate to a human intermediary to avoid being blamed. Here, our first research question is: “*Do users try to shift the blame to the programmer behind the algorithm?*”. The issue is that an algorithm itself has no personal stake in monetary losses, cannot be sent to prison, and cannot regret its actions. Thus, it cannot be held responsible for its decisions (Martin, 2019; Coeckelbergh, 2020). Therefore, either the user, the programmer, or both, could be held responsible instead of the algorithm. In light of this uncertainty, we must ask: “*Who is perceived as responsible for an inequalitarian algorithmic decision: the programmer or the user?*”. Delegating the decision is not advisable if the user believes that the programmer will not be perceived as responsible.

¹Here are some other examples of algorithms: matching algorithms are used to identify compatible profiles based on preferences (Manlove, 2013), hire employees (Dietvorst et al., 2015; Lee, 2018; Van Esch et al., 2019; Bigman et al., 2023; Dargnies et al., 2024), admit candidates to colleges (Kleinberg et al., 2018), and dismiss employees (Bartosiak and Modlinski, 2022; Corgnet, 2023). Algorithms also help detect spam, saving millions annually (Rao and Reiley, 2012), assess credit scores for banks (Lessmann et al., 2015), and detect deception through facial expressions (von Schenk et al., 2022; Leib et al., 2024). Public policies, too, use algorithms to optimize resource allocation and citizen utility (Meijer and Wessels, 2019; Papachristou et al., 2024; Chugunova and Luhan, 2024) and to manage intergenerational redistribution (Klockmann et al., 2022).

It is important to note that programmers can have different levels of influence on the outcome of an algorithm. As a result, they should not be perceived with the same degree of responsibility. A Rule-Based Algorithm (RA) is fundamentally different from an Artificial Intelligence Algorithm (AI). The RA follows logical rules created by a programmer, usually based on expert decisions. For example, MYCIN is one of the earliest rule-based expert algorithms used in medicine. Created in the 1970s, it consists of nearly 600 logical rules that allow it to make diagnoses and predict which drugs to prescribe.² Unlike RA, AIs make decisions based on a database and can work autonomously. A well known example is ChatGPT, a chat-bot developed by OpenAI. This algorithm makes predictions based on a database containing information from the internet up to a certain date.³

The processes behind AI and RA differ significantly. Unlike RA systems, where the programmer maintains complete control over the algorithmic decisions, AI operates with significantly greater autonomy. For example, in January 2023, the creator of *Game of Thrones* sued OpenAI for copyright infringement after ChatGPT reproduced sections of his work, despite OpenAI's policies against violating copyright laws ([Gerken and McMahon, 2023](#)). Similarly, Microsoft's Twitter chat-bot Tay had to be shut down just eight hours after its launch in the U.S. due to it adopting offensive behaviors influenced by user interactions, a problem not seen in China, likely due to stricter social media censorship ([Goolsbee, 2018](#)). In both cases, if the algorithms had been RA rather than AI, the programmers - and their organizations - would have had more control over the outcomes, potentially avoiding these problems. By generating its own rules, AI reduces the programmer's control and makes decisions less predictable.

The unpredictability of AI outcomes makes it difficult to determine whether the algorithm is following the programmer's intentions or is acting independently. This ambiguity inherent in AI provides programmers with the opportunity to exploit moral wiggle room ([Dana et al., 2007](#)). By making an inegalitarian decision, the AI programmer increases the likelihood that an inegalitarian allocation is selected by the AI. This brings us to our

²Others RAs exist in medicine, such as CADUCEUS and DxPlain. In finance, notable examples include Charles River Investment Management Solution and AML. In law, Thomson Reuters offers several products, including UltraTax CS, Practical Law, and Westlaw Edge.

³Currently, for ChatGPT 3.5 and 4, the database includes information from the internet up to September 2021.

last research question: *“Does an AI programmer exploit the moral wiggle room afforded by an AI?”*

To answer these questions, we use the experimental design outlined by [Bartling and Fischbacher \(2012\)](#). This design proposes a simple measure of perceived responsibility based on monetary punishment. In this one-shot economic game, a group of four participants, designated as A (the “dictator”), B (the “intermediary”), and C and D (the “recipients”), participate in the allocation of a given endowment. The participant A faces a tripartite choice: an egalitarian distribution, an inegalitarian distribution, or delegating this decision to an intermediary B. If A delegates, B must choose between the two distributions proposed to A (egalitarian or inegalitarian). After that the allocation is made, two recipients may choose to administer punishment depending on the choice made by A and B. This can be done by sacrificing a portion of their respective payoffs. This experimental design allows us to observe the perception of responsibility through monetary punishment. We introduce two new treatments, in which A can delegate decision-making to an algorithm under the control of B. In these treatments, participant A is referred to as “the dictator” or “the user”, who can delegate the decision to the algorithm. Participant B is referred to as “the intermediary” or “the programmer” who developed the algorithm. In the “RA” treatment, B has full control over the algorithm. In the “AI” treatment, the programmer’s control over the algorithm’s decisions is reduced to 50%, meaning that there is a 50% chance that the decision of participant B will be applied by the algorithm. In the other 50%, the algorithm makes the decision autonomously, based on its database.

This research primarily contributes to the literature on algorithm delegation, perceived responsibility, and moral wiggle room.

To the best of our knowledge, this paper is the first to consider the role of the algorithm programmer into an economic game. This paves the way for new research that will examine the role of the programmer in depth. (i) We show that individuals are equally likely to delegate an allocation decision to a human or an algorithm (AI, RA) when there is a programmer linked to the algorithm. (ii) We validate the theory developed by [Matthias \(2004\)](#) highlighting that the AI programmer is perceived as less respon-

sible than the RA programmer when the algorithm makes an inequalitarian allocation decision. (iii) Recipients are less likely to pay €1 to punish other participants when the intermediary is an AI. However, recipients who decide to pay €1 punish AI users more severely than RA users. This opposite effect leads AI users to receive a similar average punishment and payoff as those who delegate to a human or an RA. Thus, regardless of the nature of the intermediary, delegators still have an incentive to delegate in order to be perceived as less responsible. (iv) AI programmer exploits the moral wiggle room and therefore makes more inequalitarian choices.

We proceed as follows: In Section 2, we review the literature, then introduce our behavioral conjectures. In Section 3, we explain the framework of our experimental design and process. The results are presented in Section 4, and we conclude and discuss them in Section 5.

2 Related literature and conjectures

2.1 Related literature

To investigate whether participants decide to delegate an allocation decision to an RA or an AI, and to provide evidence of differences between these algorithms and a human intermediary, we review the relevant literature on algorithmic delegation, perceived responsibility, and moral wiggle room. We use this literature review and logical reasoning to formulate behavioral conjectures in the second part.

2.1.1 Algorithmic appreciation

Although algorithms often outperform humans in many tasks, people tend to prefer to delegate tasks to humans rather than algorithms (Meehl, 1954; Dawes, 2008; Sanders and Manrodt, 2003; Fildes and Goodwin, 2007; Dietvorst et al., 2015; Jung and Seiter, 2021; Dargnies et al., 2024). However, a preference for algorithmic delegation may emerge depending on whether the algorithm acts as an advisor (Logg et al., 2019; You et al., 2022; Daschner and Obermaier, 2022), the objectivity of the task (Castelo et al., 2019; Hertz and Wiese, 2019) and the perceived complexity of the algorithm (Lehmann et al., 2022; Chevrier et al., 2024). These findings underscore the growing importance of algorithms

in a variety of areas in economics.⁴ However, research on algorithmic responsibility and delegation remains limited. [Gazit et al. \(2023\)](#) found that humans are perceived as more responsible than algorithms in financial and medical advice, preferring human advisors because other humans can be held accountable. [Kirchkamp and Strobel \(2019\)](#) observed that humans and algorithms can be perceived as equally responsible in joint decisions leading to unequal outcomes.

2.1.2 Switching the blame to an algorithm

Prior literature suggests that principals strategically delegate decisions to agents to reduce their perceived responsibility, a phenomenon referred to as strategic delegation ([Sengul et al., 2012](#)). This allows principals to maximize payoffs ([Fershtman and Gneezy, 2001](#); [Sutan and Vranceanu, 2016](#)), maintain their social image ([Bénabou and Tirole, 2005](#); [Hamman et al., 2010](#)), or shift the blame ([Coffman, 2011](#); [Bartling and Fischbacher, 2012](#); [Oexl and Grossman, 2013](#); [Argenton et al., 2023](#)). These practices can be traced back to Machiavellian principles, where unpopular measures are delegated to others while leaders retain the means of distributing favors ([Gawn and Innes, 2021](#)).

In this context, this study examines whether individuals also delegate tasks to algorithms in order to reduce their perceived responsibility. [Feier et al. \(2022\)](#) conducted an experiment in which principals could delegate tasks to either a human or an artificial agent. The results show that participants are indifferent between delegating to both agents. However, when mistakes occur, principals who delegate to human agents are perceived as more responsible than those who delegate to artificial agents. [Awad et al. \(2020\)](#) found that users of autonomous vehicles are held responsible for accidents, rather than the companies that produce the faulty algorithms. This finding contrasts with the results of [Feier et al. \(2022\)](#), possibly due to the framing of the algorithms. Indeed, [Hou and Jung \(2021\)](#) showed that when algorithms were framed as “AI”, participants tended to value them more, whereas framing them as following logical rules increased algorithm aversion. This discrepancy in framing may explain the differences between studies.

We believe that individual perceive algorithms - RA and AI - differently, depending on how much control the programmer has. This is why we compare these two types of

⁴For a more comprehensive review, see [Jussupow et al. \(2020\)](#); [Burton et al. \(2020\)](#); [Chugunova and Sele \(2022\)](#); [Mahmud et al. \(2022\)](#).

algorithms in this paper.

2.1.3 Perceived responsibility: RA vs. AI

[Matthias \(2004\)](#) argues that it is not always possible to establish a direct link between the programmer’s decision and an algorithm’s output, especially in AI. Programmers can choose AI models and databases, but their influence is limited. If an RA makes an inequitable decision, it is a direct result of the programmer’s conscious choices. In contrast, when an AI makes an inequalitarian decision, the outcome may be the result of either the programmer’s input or biases present in the underlying database ([Zhang et al., 2020](#)). AI databases, often reflecting societal biases, have produced sexist and racist outcomes ([Rahwan et al., 2019](#); [Cowgill and Tucker, 2019](#); [Kassam and Marino, 2022](#)), and individuals can manipulate AI to their advantage ([Klockmann et al., 2022](#); [Cao et al., 2023](#)).

This ambiguity, often referred to as the “responsibility gap” ([Matthias, 2004](#)), makes it difficult to assign responsibility for decisions made by AI. Because RA programmers retain more control over the decisions of their algorithms, the causal link between the programmer’s actions and outcomes is stronger than in AI, where the algorithm can make autonomous decisions beyond the programmer’s direct input ([May, 1992](#); [Fischer, 1999](#); [Lipinski et al., 2002](#)).

Previous research suggests that AI programmers may exploit ambiguity surrounding AI outcomes to manipulate AI to produce more inequalitarian decision.

2.1.4 AI and moral wiggle room

Moral wiggle room refers to the phenomenon where individuals exploit ambiguity or uncertainty to justify self-serving decisions. By using ambiguity, individuals can make choices under the guise of uncertainty, often leading to more inequalitarian decisions. For example, [Dana et al. \(2007\)](#) has shown that in a dictator game, dictators were more likely to choose an inequalitarian distribution when the recipient was unaware of the dictator’s decision. [Larson and Capra \(2009\)](#), [Feiler \(2014\)](#) and [Grossman and Van Der Weele \(2017\)](#) conducted control checks and expanded [Dana et al. \(2007\)](#)’s design, showing that participants preferred to remain ignorant about their actions’ effects on other participants’ payoffs. Moral wiggle room allows participants to avoid information about the

consequences of their actions. All of these studies show that moral wiggle room enables participants to make more self-serving decisions. Based on this logic, [Garcia et al. \(2020\)](#) showed that participants act strategically by avoiding information about others' payoffs as an excuse to be less generous in charitable giving without feeling guilty.

As we have seen, exploiting moral wiggle room is well known in the literature. To the best of our knowledge, no paper has yet demonstrated that, due to the ambiguous outcomes delivered by AI, participants may also exploit moral wiggle room to make more inequalitarian allocation decisions.

2.2 Behavioral conjectures

Our behavioral conjectures are based on the idea that participants can delegate an allocation decision to an intermediary. The intermediary can be a human, an RA, or an AI. Recipients can also decide to punish another participant by sacrificing a portion of their respective payoff to reduce another participant's payoff. We thus distinguish two variables: the likelihood of being punished, which is reflected in the recipient's decision to sacrifice their payoff. And the severity of the punishment, which is captured by the amount of euros deducted from other participants (the design is described in detail in the next section).

2.2.1 Delegation

In this game, if all participants are rational and want to maximize their payoff, the dictator will choose the inequalitarian decision directly, and no recipient will choose to sacrifice their payoff to punish. However, several studies have shown that the empirical results do not match the theory ([Fehr and Schmidt, 2006](#)). Thus, the dictator may anticipate a possible punishment. Strategically, if the dictator assumes that the intermediary is driven by the objective of maximizing profit and the recipient has a preference for fairness and will punish, the dictator should delegate to the intermediary to let him/her be punished instead. For the "switching the blame" strategy to be effective, the intermediary must be perceived as responsible ([Bartling and Fischbacher, 2012](#); [Argenton et al., 2023](#)). Previous studies, similar to our baseline, have already shown that the "switching the blame" strategy is used. In our RA treatment, participants are well aware that the programmer

completely controls the decisions made by the algorithm. The only difference between the *Human* and the *RA* treatment is the framing. In the *RA* treatment, we say that the “algorithm has selected” instead of “another participant has selected” a certain allocation. However, it is always the programmer who makes all the algorithm’s decisions and can be held accountable. Participants know this, so we expect no difference.⁵ Regarding the *AI* treatment, we assume that the *AI* programmer will be perceived as less responsible because the algorithm can make decisions autonomously. The question is whether the dictator will anticipate this. In [Bartling and Fischbacher \(2012\)](#), a similar experiment was conducted with delegation to a die. Behind the die, a participant assumed responsibility for the random draw. There was no difference in the number of delegations between delegating to another human and delegating to the die. [Feier et al. \(2022\)](#) found no significant difference in the tendency to delegate tasks between humans and algorithms. This suggests that individuals who delegate tasks do not anticipate that algorithms cannot assume responsibility in the same way as humans. They assume that participants cannot anticipate that the artificial agent will be less punished.⁶ Based on these two papers, we assume that the dictator will not anticipate that the *AI* intermediary will be less blamed and that the blame will fall on them.

Conjecture 1: *Participants’ delegation decisions are not influenced by the type of intermediary (human, RA, or AI).*

2.2.2 Perceived responsibility

[May \(1992\)](#) and [Lipinski et al. \(2002\)](#) showed that for an individual to be judged responsible, it is necessary for the judge to see the causal link between an individual decision and the resulting outcome. The stronger the causal link between making the decision and the outcome, the more participants can judge the decision maker as being responsible for their actions. If there is a delegation to an intermediary, such as another participant or an *RA*, the direct link between the outcome and the decision-maker is quite easy to draw. The recipients should perceive the intermediary as responsible ([Bartling and Fischbacher, 2012](#); [Oexl and Grossman, 2013](#)). In the case of *AI*, due to the ambiguity

⁵As mentioned in the design section, we implement the *RA* treatment to isolate the effect of the algorithmic framing.

⁶“However, when negative outcomes occurred, delegators fared significantly better if a machine agent caused the failure. Interestingly, participants did not seem to anticipate this pattern as we did not find significant differences regarding delegation decisions themselves.” [Feier et al. \(2022\)](#)

about who makes the choice (the AI programmer or the AI), the direct link between the AI programmer and the AI decision is less strong. This main difference between RA and AI is the key factor to understand our following conjectures.

[Bartling and Fischbacher \(2012\)](#) used a punishment pattern benchmark to determine who is perceived as responsible for the outcome. The harsher a participant is punished, the more responsible they are perceived. To administer punishment, the recipient must pay to be allowed to reduce another participant's payoff. Recipients who have a preference for fairness and are willing to pay to punish others should be sure about who is truly responsible for the decision. The risk of punishing an innocent participant emerges in the *AI* condition. Therefore, the willingness of the recipients to punish should diminish due to this ambiguity. Recipients should thus less frequently pay to punish others in the AI treatment because they have difficulty determining who should be held responsible in this treatment condition. Due to the lack of control that AI programmers have over the AI decision, our conjecture [2.1](#) highlights that AI programmers should be perceived as less responsible than RA programmers ([Matthias, 2004](#)).

Conjecture 2: *It is less likely that recipients will decide to punish other participants when an inequalitarian allocation is selected by an AI.*

AI programmers have much less control over the algorithm's decision-making, which results in them being perceived as less responsible than RA programmers who directly determine the algorithm's decisions. The direct link between the decision made by an RA and its programmer is stronger than the direct link between the decision made by an AI and its programmer. The AI makes an autonomous decision without considering the programmer in 50% of cases. Therefore, we assume that participants will perceive an RA programmer as more responsible for the decision-making by the algorithm than an AI programmer, who cannot completely control it. According to [Bartling and Fischbacher \(2012\)](#), being perceived as more responsible leads to harsher punishment. Therefore, RA programmer punishment should be harsher than AI programmer punishment.

Those who are willing to pay should perceive the AI user as more responsible, given that the AI user has fully decided to delegate to the AI rather than making an egalitarian decision. Conjecture [2.2](#) aligns with findings from [Awad et al. \(2020\)](#), who demonstrated

that people perceive autonomous vehicle users as more responsible than the firm that produces them. [Bartling and Fischbacher \(2012\)](#) demonstrated that when an individual delegates a decision to a dice roll, she is perceived as significantly more responsible for the outcome than the mechanism itself. Similarly, [Oexl and Grossman \(2013\)](#) conducted an experiment in which the intermediary is permitted to select only an inequalitarian allocation. When the dictator delegates to an intermediary who is compelled to make an inequalitarian choice, the recipient perceives the dictator who delegates to the intermediary as more responsible. In a firing context, [Corgnet \(2023\)](#) showed that a manager is perceived more negatively by the dismissed worker when the manager delegates the decision to an algorithm than in the absence of delegation. These papers highlight that when intermediaries are limited in their decision-making abilities or lack control over the decision, the individual delegating the decision is perceived as responsible for the outcome. Thus, we assume that the AI user will be punished more harshly than the RA user.

Conjecture 2.1: *It is more likely that an RA programmer will be punished more severely for an inequalitarian decision made by the algorithm than an AI programmer.*

Conjecture 2.2: *It is more likely that an AI users will be punished more severely for an inequalitarian decision made by the algorithm than an RA users.*

According to our conjectures [2](#) and [2.1](#), the AI programmer will be less blamed and punished less severely than the RA programmer. Therefore, the expected utility of an AI programmer should be higher than that of an RA programmer if the algorithm selects the inequalitarian allocation. Concerning the AI user, there is a confounding factor due to conjectures [2](#), and [2.2](#). The AI user will be punished less often, but the punishment will be more severe when it occurs. Depending on which effect is stronger, the AI user may receive a monetary payment equal to or different from that of the RA user. We cannot predict which of these opposing effects will be stronger. Therefore we cannot determine whether delegating to an AI can really protect the user from punishment.

2.3 AI, moral wiggle room and welfare

Finally, conjecture [2.1](#) suggests that AI programmers are less likely to be punished when the AI algorithm selects the inequalitarian allocation. Conjecture [2](#) indicates that par-

ticipants from *AI* treatment are less frequently subjected to punishment. Thus, it is possible for AI programmers to exploit the moral wiggle room left by the AI’s decision. For example, [Dana et al. \(2007\)](#) used a dictator game with two choices, egalitarian and inequalitarian. They showed that when the choice is certain, 26% of the dictators choose the inequitable choice. But when the choice is ambiguous, allowing the use of moral wiggle room, 63% of them choose the inequitable option. Similarly, it is possible that AI programmers use the moral wiggle room to select the inequalitarian allocation because they assume that they will be perceived as less responsible, and thus be punished less harshly.

Conjecture 3: *AI programmers are more likely to select the inequalitarian allocation compared to the human intermediary and the RA programmer.*

3 Design

Based on the literature, we designed an experiment that allows us to measure the perceived responsibility for an inequalitarian decision made by different types of intermediaries. It also provides information about the decisions made by the participants, allowing us to answer our research questions.

3.1 Human intermediary

The baseline of this study is a derivative of the designs of [Bartling and Fischbacher \(2012\)](#). This game involves the formation of a group of four participants, designated as A (the “dictator”), B (the “intermediary”), and C and D (the “recipients”), engaged in the division of a €20 endowment. The baseline is referred to as the *Human* treatment.

Participant A faces a choice between two different allocations: an egalitarian allocation, where each participant receives €5, and an inequalitarian allocation, where participants A and B receive €9 and participants C and D €1. A key aspect of our design is that participant A can delegate the decision-making to the intermediary, B, who then faces the same allocation choices. Regardless of participant A’s decision, participant B has to choose between an egalitarian and an inequalitarian distribution.⁷ Meanwhile, partic-

⁷Compared to the original design, we decided to use the strategy method for the decisions made by the intermediary participant (B) in order to obtain more observations and answer our third research question.

ipants C and D express their decisions in all possible situations. This step allows us to elicit 4 distinct situations depending on the choices made by A and B, with respect to the allocation type.⁸ In each situation, participants C and D have the possibility to punish other participants by spending €1 of their own allocation. They can then reduce other participants payments by up to €7 in total. The recipients choose the amount they want to deduct from participants A, B, and/or the other recipient.⁹ The four potential situations faced by C and D occur in a random order. At the end of the experiment, one recipient - either C or D - is randomly selected, and one situation is chosen based on the choices made by A and B. The non-selected recipient does not incur an expenditure cost of €1. This punitive mechanism allows for the elicitation of the responsibility perceived by C and D towards A and B (Bartling and Fischbacher, 2012).

3.2 RA and AI programmer

The baseline is extended through two new treatments that redefine the role of participant B as a programmer exerting control over a decision-making algorithm (see Figure 1). In the RA treatment, the RA programmer has full control over the algorithm, allowing participant A - the RA user - to either directly select an allocation or delegate the decision to the RA, fully controlled by participant B - the RA programmer. It should be noted that the sole distinction between the *Human* and the *RA* treatment is the framing of the algorithm. The AI treatment introduces a variation wherein the AI algorithm applies the AI programmer's decision in 50% of the cases. In this treatment, participant B - the AI programmer - controls the AI algorithm's decision in 50% of the cases. When participant A - the AI user - delegates to an algorithm, there is a 50% chance that the AI algorithm will apply the choice made by the AI programmer. In the remaining 50% of cases, the AI algorithm chooses autonomously. The AI algorithm is presented to participants as an algorithm that makes decisions based on previous human decision (see Appendix A). To achieve this, we calibrate our AI algorithm by conducting a pre-study.¹⁰ Participants are not informed of the distribution used by the AI algorithm to make its decision. Par-

⁸There are 4 different scenarios: A chooses egalitarian (5,5,5,5); A chooses inequalitarian (9,9,1,1); A delegates and B chooses inequalitarian (9,9,1,1); A delegates and B chooses egalitarian (5,5,5,5).

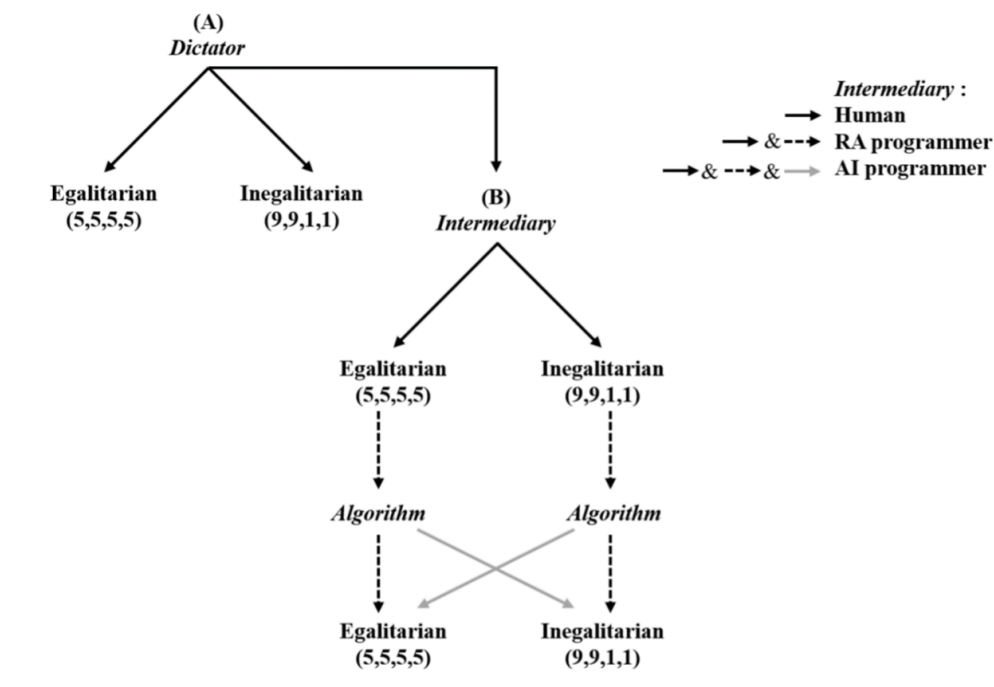
⁹The other recipient can be punished to avoid the demand effect (Zizzo, 2010).

¹⁰In this pre-study, we only formed groups of two participants. Only the roles of participants B and C were retained. The payoffs and punishments were identical to those in our main experiment. Participants B could choose between two distributions: an egalitarian allocation (5,5) or an inequalitarian allocation (9,1). Recipients C had the option to deduct up to €7 as a punishment by sacrificing €1. 48% of participants B chose the inequalitarian allocation.

Participants C and D do not know the choice of the decision made by the AI programmer. They can just observe the outcomes of the algorithm’s decisions. During the experiment, we use generic terms to refer to our participants. For all experiments, we designate them as participant A, B, C, or D, depending on their role (see Appendix A).

These three treatments were implemented using a between-subject design. To avoid any learning effect, it was a one-shot experiment.

Figure 1: Experimental design – Three decisions across treatments



“Human” treatment is represented by *the full black arrow*. “RA” treatment is represented by *the full black and dash arrow*. “AI” treatment is represented by *the full black, dash, and full grey arrow*. In all treatments, the Dictator (A) can choose between 2 allocations or can delegate to the intermediary. Egalitarian allocation means that all players earn €5. Inegalitarian allocation means that the dictator and the intermediary earn €9, while recipients C and D earn €1. If the dictator delegates, the decision taken by the intermediary is retained. In the “RA” and “AI” treatments, the intermediary is respectively the RA programmer and the AI programmer. The “RA” programmer controls the decision applied by the algorithm. The “AI” programmer has a 50% chance that the algorithm applies his decision and a 50% chance that the AI algorithm makes the decision itself. If the AI algorithm makes the decision itself, there is a 48% chance that it selects the inegalitarian decision and a 52% chance that it selects the egalitarian decision.

3.3 Experimental process

We used the ORSEE database (Greiner, 2015) of the Laboratory of Experimental Economics of Nice (LEEN) to recruit participants for both our pre-study and main study. The experiments were conducted at the LEEN. The experimental tasks were programmed

using the oTree software (Chen et al., 2016) and deployed through the Heroku cloud platform.¹¹

Upon arrival at LEEN, participants were randomly assigned to a seat by drawing a ball from an urn to eliminate any potential role allocation bias within the experiment. This initial step was followed by a comprehensive oral briefing of the experimental instructions, delivered by the same experimenter. Participants were then subjected to a comprehension test to ensure a clear understanding of the experimental procedure. Those who were unable to demonstrate understanding of the experiment’s rules, as evidenced by more than three errors in the comprehension test, were excluded from participation.¹²

Participants were compensated with a show-up fee of €5. The main game of the experiment yielding an additional average earning of €4.59, resulting in an average total remuneration of €9.59 per participant. The pre-study sessions were more concise due to the reduced group size, lasting approximately 15 minutes. The main experimental sessions lasted approximately 30 minutes.

A component of our methodology was the calibration of the AI algorithm, achieved through a pre-study conducted in December 2023 with 46 participants. This preliminary phase was mandatory to calibrate the AI algorithm. The main study took place in January 2024. Participants who took part in the pre-study were excluded to avoid contamination of the results. Each treatment involved 120 participants, for a total of 360 participants across all treatments, spread over 25 experimental sessions.

Upon the completion of the main experimental game, participants completed a post-experiment questionnaire. This session included an adapted version of the Cognitive Reflection Test, based on the works of Frederick (2005), Toplak et al. (2014), and Primi et al. (2016), to assess participants’ cognitive processing and decision-making styles. Additionally, the interaction scale of the Negative Attitudes toward Robots Scale (NARS) (Nomura et al., 2006; Dinet and Vivian, 2014), risk preference scale (Dohmen et al., 2011), and the Control Attitude Scale (Wolff et al., 2022) were administered to glean

¹¹The study was pre-registered on As Predicted at this link: <https://aspredicted.org/vi74k.pdf>.

¹²We excluded 11 participants out of 371.

insights into participants' predispositions towards robot and their perceptions of risk, ambiguity and control.¹³

4 Results

4.1 Dependent variables

Our results section is divided into four parts. First, we compare the likelihood that a dictator delegates the decision to an intermediary using a proportion test. Then, we compare the proportion of times that recipients paid to punish across treatments and how they allocated punishment within their group. Wilcoxon rank-sum tests are used to compare patterns of punishment.¹⁴ To control for individual factors, we conduct an OLS regression. Intermediary preferences are analyzed in our third section using a proportion test and probit analysis.

4.2 Delegation

Table 1 shows the percentage of times participant A chooses a particular option per treatment. For *Human* and *RA programmer* treatments, participants A demonstrate an identical choice distribution. No significant difference with a proportion test is observed between *Human* (or *RA*)¹⁵ and *AI* in terms of inequalitarian allocation (20% vs 33%, $\chi^2 = 0.76$, $p = 0.38$), egalitarian allocation (57% vs. 50%, $\chi^2 = 0.06$, $p = 0.79$), and delegation (23% vs 17%, $\chi^2 = 0.10$, $p = 0.74$). However, compared to the findings of Bartling and Fischbacher (2012), our replication indicates that participants are less likely to delegate decisions and more inclined towards choosing the egalitarian option.

The likelihood that the dictator decides to delegate is not different between treatments. Indeed, the decision to delegate remains the same, which suggests that in every case, the expectation of participant A regarding the ability of the intermediary to take the blame is similar. Participants A's decisions are not sensitive to the type of algorithm they can delegate to. Bartling and Fischbacher (2012) observe a similar result in their die treatment, where participants continue to delegate the decision to a die. Feier et al. (2022)

¹³All experimental materials are available in the Appendix and described in Table 5.

¹⁴We only report the results of the Wilcoxon signed rank test, but the t-tests give a similar result.

¹⁵For the sake of simplicity, we did not report the p-values for *Human* vs *RA* programmer as the distributions are identical (see Table 1).

elicit the same tendency by showing that between human and artificial treatments, the likelihood that the agent delegates the task to an artificial or human agent is similar.

These results support Conjecture 1. Indeed, we observe no difference in delegation between the treatments.¹⁶

Result 1: The decision to delegate is not influenced by the type of intermediary (human or algorithm) nor the type of algorithm (RA or AI).

Table 1: Delegation and allocation choices of dictator per conditions.

	A inegalitarian (%)	A egalitarian (%)	A delegation (%)
Human	33	50	17
RA programmer	33	50	17
AI programmer	20	57	23

“A” is the dictator; “Human” means the intermediary is a human; “RA programmer” means the intermediary is an algorithm controlled at 100% by participant B; “AI programmer” means the intermediary is an algorithm controlled at 50% by the participant.

4.3 Pay to Punish and Punishment Pattern

In the column “obs” of Table 2, the proportion of participants C and D that decide to pay €1 to punish another participant is not statistically different when the allocation decision selected is egalitarian.¹⁷ When player A delegates the decision to an AI that makes an inegalitarian decision, participants C and D choose to pay €1 to punish in 36% of cases. Recipients are much more likely to pay €1 to punish when player A delegates to an RA or a human, resulting in an inegalitarian allocation, according to the proportion test (AI = 36% vs RA = 63%, $\chi^2 = 7.5$, $p = 0.006$), (AI = 36% vs Hum. = 58%, $\chi^2 = 7.5$, $p = 0.02$). These results support Conjecture 2, the likelihood of being punished is lower when the decision is made by an AI rather than an RA. This result is explained by the ambiguity left by the AI. Since the participants do not know whether the result was generated by participant B or by the AI, they do not know who should be held

¹⁶This is confirmed by our probit regressions in Appendix 6

¹⁷Statistical differences across treatments when the egalitarian decision is selected without delegation (See “Obs” value in Table 2) : (AI = 1% vs RA = 11%, $\chi^2 = 3.3482$, $p = 0.07$); (AI = 1% vs Hum. = 5%, $\chi^2 = 0.2586$, $p = 0.61$); (RA = 11% vs Hum. = 5%, $\chi^2 = 0.9818$, $p = 0.32$), when the egalitarian decision is selected with delegation (AI = 5% vs RA = 8%, $\chi^2 = 0.13393$, $p = 0.7144$); (AI = 5% vs Hum. = 6%, $\chi^2 < 0.00001$, $p = 1$); (RA = 8% vs Hum. = 6%, $\chi^2 < 0.00001$, $p = 1$).

responsible and decide not to pay €1.

Result 2: Delegating the task to an AI reduces the chance of being punished.

In the following analysis, we will focus on punishment patterns in inequalitarian decisions as shown in Table 2 and Figure 2. We start the first subsection with the analysis of the programmer, followed by user responsibility.

Table 2: Punishment pattern and Number of punishment observation per treatment.

Punishment on:	A inequalitarian				B inequalitarian			
	Obs	A	B	C D	Obs	A	B	C D
Human	32	5.00	1.56	0.06	35	1.77	4.54	0.14
RA programmer	34	4.65	2.00	0.12	38	2.21	4.45	0.13
AI programmer	31	5.03	1.71	0.00	22	3.91	2.50	0.14

Punishment on:	A egalitarian				B egalitarian			
	Obs	A	B	C D	Obs	A	B	C D
Human	3	1.67	2.33	0.00	4	2.00	3.00	1.50
RA programmer	7	2.14	2.14	0.71	5	1.40	2.60	0.60
AI programmer	1	2.00	2.00	3.00	3	2.00	1.67	0.00

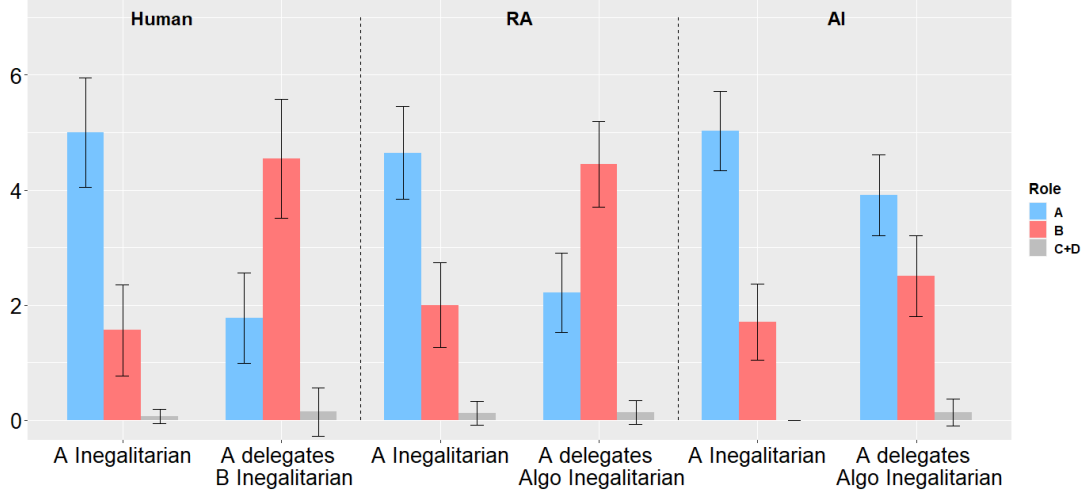
“Obs” refers to the number of times that participants C and D decide to punish depending on the situation. 60 participants C and D were Participant A is “the dictator”, participant B is “the intermediary”, and participant C/D are “the other recipient”. The upper table shows situations where A and B (when A delegates), select the inequalitarian decision. The lower table shows situations where A and B (when A delegates), selecting the egalitarian decision. “Human” means the intermediary is B (a human); “RA programmer” means the intermediary is an algorithm fully controlled by participant B; “AI programmer” means the intermediary is an algorithm controlled at 50% by the participant.

4.3.1 Punishment pattern: programmer

As shown in Table 2 and Figure 2, the punishment distributions follow the same tendency between our *Human* and *RA* treatments. The punishment level of participant A is not significantly different between *Human* and *RA* treatments.¹⁸ Thus, programmers who perfectly control the algorithm’s allocation decisions are perceived as responsible, as if they had made the decision themselves.

¹⁸When participant A chooses an inequalitarian allocation (Hum. = €5.00 vs RA = €4.64, Rank Sum Test, p-value = 0.47), an egalitarian allocation (Hum. = €0.06 vs RA = €0.20, Rank Sum Test, p-value = 0.64), and delegates the decision, participant B chooses an inequalitarian allocation (Hum. = €1.77 vs RA = €2.21, Rank Sum Test, p-value = 0.10) and an egalitarian allocation (Hum. = €0.11 vs RA = €0.10, Rank Sum Test, p-value = 0.73)

Figure 2: Punishment pattern per treatment when an inequalitarian allocation is selected and participants C and D decide to punish.



“A” is the dictator, “B” is the intermediary; “Human” means the intermediary is B (a human); “RA programmer” means the intermediary is an algorithm fully controlled by participant B; “AI programmer” means the intermediary is an algorithm controlled at 50% by the participant. The probability that a participants C and D decides to punish when player A (or when player A delegates and player B) chooses an inequalitarian decision is 53% (58%) for a human, 56% (63%) for a rule-based algorithm (RA), and 51% (36%) for an AI.

Nevertheless, an algorithm that selects an inequalitarian distribution results in a significantly lower punishment for the AI programmer compared to the RA programmer (AI = €2.50 vs RA = €4.45, Rank Sum Test, $W = 170$, $p < 0.00001$) (Figure 2). The AI programmer is perceived as less responsible than the AI user (AI: Prog. = €2.50 vs User = €3.90, Rank Sum Test, $W = 364$, $p = 0.003$), and the RA programmer is perceived as more responsible than the RA user (RA: Prog. = €4.45 vs User = €2.21, Rank Sum Test, $W = 187$, $p < 0.00001$).

Table 4 uses ordinary least squares (columns 1 to 4) to analyze the pattern of punishment chosen by participants C and D. Columns (1) and (2) represent the number of punishment points received by the principal and columns (3) and (4) represent the punishment received by the intermediary. Our OLS regressions give support to conjectures 2, and 2.1. Indeed, regressions (3) and (4) of Table 4 report a significant reduction in the amount of punishment for the AI programmer treatment (coef = -0.868, $p < 0.001$).¹⁹

This finding aligns with the “responsibility gap” theorized by Matthias (2004), and our

¹⁹With control for individual factors (coef = -0.759, $p < 0.001$).

Conjecture 2.1 according to which the AI programmers will be perceived as less responsible than the RA programmers.

Result 2.1: The AI programmer is punished less severely than the RA programmer.

4.3.2 Punishment pattern: user

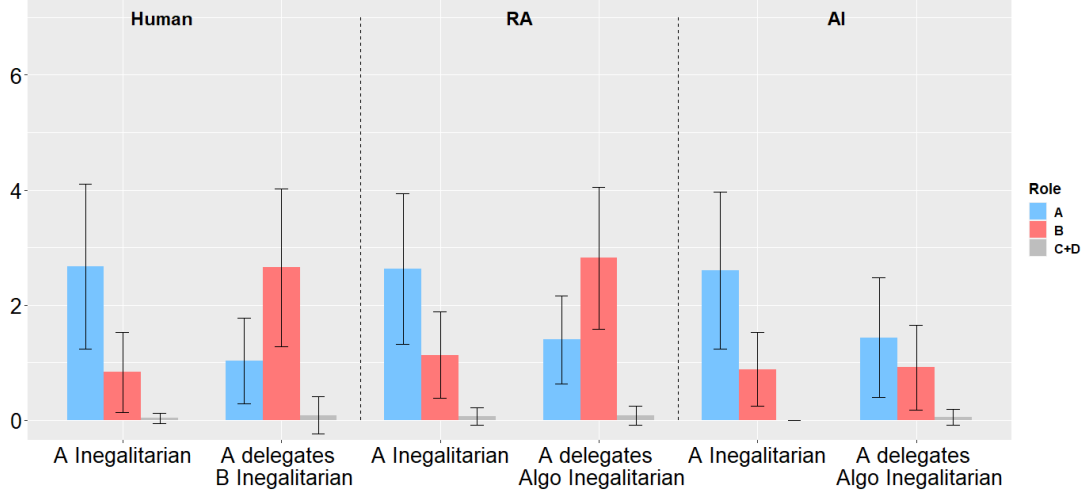
As indicated by Figure 2, when the user delegates the allocation decision to an algorithm that selects the inequalitarian allocation, the participant A (user) from *AI* condition is punished more harshly than the participant A (user) from *RA* condition (Users : AI = 3.91 vs. RA = 2.21, Rank Sum Test, $W = 692.5$, $p < 0.00001$). Regressions (1) and (2) of Table 4 also support our Conjecture 2.2. The AI user is significantly more severely punished in our AI Programmer treatment both with and without controls (coef= 0.938, $p < 0.001$; coef=0.879, $p < 0.001$).

Result 2.2: The AI user is punished more severely than the RA users.

According to previous results 2, 2.1, and 2.2, participants A are less punished if they delegate the allocation decision to a human or an RA who selects the inequalitarian allocation, than by selecting this allocation themselves. Participant A, who delegates the allocation decision to an AI that makes the inequalitarian decision, is less likely to be blamed, but if they are blamed, the punishment is harsher than in the *Human* or *RA* treatments. To construct Figure 3, we consider that if participants C and D chose not to punish, the amount of punishment is equal to 0. We thus obtain the expected punishment. In the event that Participant A delegates the decision to an AI or an RA, the expected punishment is similar if the selected decision is inequalitarian (Users : AI = €1.43 vs RA = €1.40, Rank Sum Test, $W = 1880$, $p = 0.6431$). Participant A's expected punishment is lower when delegating to an AI than when selecting the inequalitarian decision themselves (AI: user = €1.43 vs prog. = €2.60, Rank Sum Test, $W = 2214$, p-value = 0.01). Thus, delegating a decision to an AI allows participant A to be perceived as less responsible for the algorithm's actions, similar to the RA or human treatment.

Result 2.3: Delegating the decision results in the delegator being perceived as less responsible.

Figure 3: Expected Punishment pattern per treatment when an Inegalitarian allocation is selected.



“A” is the dictator, “B” is the intermediary; “Human” means the intermediary is B (a human); “RA programmer” means the intermediary is an algorithm fully controlled by participant B; “AI programmer” means the intermediary is an algorithm controlled at 50% by the participant. Participants C and D, who decide not to pay €1 to punish, are recorded in the table as participants who give €0 of monetary punishment.

4.4 AI and moral wiggle room

The introduction of the algorithm in the *RA* treatment does not cause a deviation in *RA* programmer decision-making from the *Human* treatment as shown in Table 3. This is supported by our probit regressions in Table 4 columns (5) and (6). In these regressions, the variable of interest equals 1 if player B chooses the inegalitarian outcome and 0 otherwise. However, the *AI* programmer increases the probability of choosing the inegalitarian allocation (60%), compared to *RA* programmer (47%) and the intermediary in our *Human* treatment (30%) (Table 3). The proportion test shows a difference between *AI* and *Human* treatments in the decision made by B, i.e the programmer ($AI = 60\%$ vs $Hum. = 30\%$, Proportion Test, $\chi^2 = 3.28$, $p = 0.03$). Furthermore, our probit regressions (5) and (6) of table 4 confirm the robustness of this result (with and without the control variable), as the probability of an inegalitarian decision increases in the *AI* treatment

(coef= 0.684, $p < 0.05$; coef=0.844, $p < 0.05$). These results confirm Conjecture 3, which shows that the AI programmer selects the inequalitarian distribution significantly more often than the human intermediary.

Result 3: AI programmer exploits the moral wiggle room and selects the inequalitarian decision more frequently.

Table 3: Allocation choices of the intermediary by treatment.

	B inequalitarian (%)	B egalitarian (%)
Human	33	67
RA programmer	47	53
AI programmer	60	40

For the AI programmer treatment, we report the choice made by the programmer and not the outcome of the algorithm. “B” is the intermediary; “Human” means the intermediary is B (a human); “RA programmer” means the intermediary is an algorithm fully controlled by participant B; “AI programmer” means the intermediary is an algorithm controlled at 50% by the participant.

4.5 Extended Results: Risk Attitude

These results were not expected in our pre-registration and were not initially included in our hypothesis.

Regressions (5) and (6) in Table 4 show that subjects who are more willing to take risks in their daily lives are more likely to choose the inequalitarian option. Indeed, the variable Risk is equal to 0 for a risk-averse subject and 10 for a risk-lover subject.²⁰ This finding is consistent with our design, as a risk-averse agent would fear punishment more than a risk-taking agent. Indeed, by choosing the inequalitarian distribution, they maximize their payoff before the punishment. But they also accept an increased risk of being punished by players C and D. By choosing the egalitarian outcome, they maximize the payoff of C and D, which should significantly reduce C and D’s willingness to punish. But at the same time, they accept a reduction in their pre-punishment payoff from 9 to 5. This trivially explains the significance of risk attitudes in our regressions.

²⁰For mor detail please refer to Appendix C

Result 4: Risk lover intermediaries are more likely to choose the inequalitarian distribution.

5 Discussion and conclusion

In this paper, we use the original experimental design created by [Bartling and Fischbacher \(2012\)](#) to address three research questions. (1) “*Do users try to shift the blame to the programmer behind the algorithm?*”. (2) “*Who is perceived as responsible for an inequalitarian algorithmic decision, the programmer or the user?*”. (3) “*Does an AI programmer exploit the moral wiggle room afforded by an AI?*”. To respond to these questions, we reproduce the “Delegation & Punishment” condition from [Bartling and Fischbacher \(2012\)](#) and add two new treatments where the intermediary is either an RA or an AI.

Compared to the original paper, 50% of our participants A selected the egalitarian allocation in our *Human* treatment, compared to 28% in the original paper. Similar differences are found concerning the inequalitarian allocation (33%) and the delegation (17%), against 17% for the inequalitarian allocation and 55% for delegation in the original paper. In our study, participants delegate much less compared to the original study. We may perhaps explain this by suggesting that dictators have a greater preference for control ([Donnelan and Robins, 2010](#)). Alternatively, individuals with a Cognitive Reflection Test (CRT) score lower than the average may be less inclined to engage in strategic reasoning, thus ignoring the delegation option because they fail to perceive its benefits.²¹

Due to the use of the strategy method, we are able to observe the punishment patterns that recipients are willing to apply, even if the delegation rate between [Bartling and Fischbacher \(2012\)](#) and our paper differs. In the *Human* treatment we found a similar tendency as in [Bartling and Fischbacher \(2012\)](#) concerning the punishment pattern. When participants A choose the inequalitarian allocation, they receive a punishment equivalent

²¹Participants in our study had an average CRT score ([Frederick, 2005](#)) of 0.94 (SD = 1.05), which is significantly lower than the score observed in other studies such as [Otero and Alonso \(2023\)](#), based on the average European population (n = 602), which shows a score of 1.27 (SD = 1.12) (0.94 vs 1.27, Z-test, p < 0.00001). [Stieger and Reips \(2016\)](#) conducted a study in Austria and Germany (n = 2272), which shows a significantly higher average score of 1.16 (SD = 1.08) (0.94 vs 1.16n Z-test, p < 0.00001) compared to our study. Among students with university degrees, the average score is 1.55 (SD = 1.07) (0.94 vs 1.55, Z-test, p < 0.00001). In the original paper of [Frederick \(2005\)](#), the CRT score was performed in 11 universities in the USA; the average score was about 1.24. At the top of this ranking was MIT with an average of 2.18, and at the lower end was the University of Toledo with 0.57. Those pieces of evidence indicate that, on average, our participants scored lower than average.

Table 4: Regressions for “Punishment Pattern” and “Pay to Punish” when the allocation is inequalitarian.

Dependant Variable	(1)	(2)	(3)	(4)	(5)	(6)
Player's role	Punishment Pattern				Pay to Punish	
	Principal		Intermediary			
RA Programmer	0.0503 (0.25)	0.0901 (0.50)	0.107 (0.46)	0.0979 (0.42)	0.347 (1.05)	0.657 (1.73)
AI Programmer	0.938*** (3.79)	0.879*** (3.56)	-0.868*** (-3.09)	-0.759*** (-2.62)	0.684* (2.05)	0.844* (2.17)
A egalitarian	-2.686*** (-6.54)	-2.721*** (-6.37)	0.190 (0.45)	0.240 (0.54)		
B inequalitarian	-2.364*** (-8.87)	-2.393*** (-8.86)	2.187*** (8.18)	2.202*** (8.07)		
B egalitarian	-3.075*** (-8.18)	-3.071*** (-7.51)	0.670 (1.32)	0.662 (1.37)		
Male		-0.321 (-1.36)		0.107 (0.45)		0.107 (0.33)
Risk		0.0710 (1.50)		-0.00271 (-0.05)		0.244*** (3.27)
Ambiguity		-0.00832 (-0.20)		0.00503 (0.12)		-0.0243 (-0.49)
Control		-0.00354 (-0.05)		-0.0602 (-0.92)		-0.0372 (-0.50)
CRT True		0.0962 (1.42)		-0.0178 (-0.25)		-0.0371 (-0.40)
Nars		0.00381 (0.17)		-0.0172 (-0.70)		-0.0181 (-0.57)
Control	No	Yes	No	Yes	No	Yes
Constant	4.569*** (19.93)	4.024*** (7.45)	2.003*** (9.14)	2.449*** (3.56)	-0.431 (-1.81)	-0.814 (-0.96)
R^2	0.4310	0.4504	0.3407	0.3515	0.0349	0.2329
Observations	215	215	215	215	90	90

t statistics in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

“Punishment Pattern” is a variable ranging from 0 to 7. Participants C and D allocate punishment between the Principal and the Intermediary. In our table, we control for the amount of punishment allocated to the Principal or the Intermediary. The table presents the results of OLS regressions (1) to (4) concerning the “Punishment Pattern”. “Pay to Punish” for the inequalitarian decision is a binary variable: 1 if the participant pays €1 to punish, and €0 if not. We use a probit model in regressions (5) and (6). For each regression, robust standard errors are reported in parentheses, clustered at the individual level. See C for the description of variable.

to 4.27 points in the original paper, compared to €5 in our *Human* treatment. When they choose to delegate to B, and B selects the inequalitarian choice, the same similarity appears. In the original paper, under that condition, A (B) receives 1.31 (3.96) points of punishment, while in our replication A (B) loses €1.77 (€4.54).²²

Comparing our baseline to the RA and AI conditions allows us to validate our behavioral conjecture. We do not observe any significant difference in the decision to delegate across our treatments. This indicates that the delegator is indifferent between delegating to a human or an algorithm, and between a fully controlled RA and an AI. However, when the intermediary chooses an inequalitarian allocation, different results emerge. There is no difference in the punishment decision between a human intermediary and an RA; in both cases, the intermediary is perceived more responsible than the delegator. But when an AI is introduced, this result is reversed. When the programmer loses control over the algorithm, she is perceived as less responsible, and the user who delegated the task is seen as responsible for the algorithm’s outcome. Finally, the programmer exploits the moral wiggle room provided by the AI and chooses the inequalitarian distribution more frequently.

We address a number of contributions from a variety of disciplines in order to provide a comprehensive overview. The practice of delegating a decision in order to be perceived as less responsible is a common occurrence in both corporate and political environments. In this context, we would like to introduce the idea that participants can also use an algorithm to shift responsibility. In terms of public policy, if companies are aware of this shift in perceived responsibility, it may create an opportunity for them to develop profitable algorithms regardless of the consequences for the population. This allows companies to shift responsibility for the decisions made by algorithms onto users while reaping the profits, thereby increasing inequality between users, programmers and recipients. This reinforces the need for a robust regulatory framework. In particular, legal liability laws play a crucial role in protecting individuals from such situations.

This study represents an initial exploration that can be expanded. Firstly, we posit

²²Refer to Appendix D to see [Bartling and Fischbacher \(2012\)](#) figure of punishment patterns compared to ours

a critical assumption: the AI programmer exerts 50% control over the algorithm. This theoretical assumption, however, does not reflect real-world scenarios and varies significantly among AI programmers. Prior research by [Bartling and Fischbacher \(2012\)](#) indicates that when control is absent (0%), human intermediaries are not perceived as responsible. Responsibility is fully attributed to the dictator, suggesting a decrease in responsibility proportional to the intermediary's level of control. A first extension for further research would be to investigate whether the reduction in punishment is monotonic depending on the percentage of control the programmer has on the AI. The second major assumption is that individuals are aware that AI learning algorithms work autonomously and that AI programmers have limited influence over these systems. Third, our model assumes that participant B cannot select their role, either as an AI or RA programmer. Another limitation of this laboratory experiment is that we assume students, selected from various fields, will act as programmers. A field experiment to obtain natural data could be conducted to compare lab data with field data. Another aspect that we neglect in this experiment is that a firm's strategy and the programmer's preferences may conflict. Finally, algorithms are evolving rapidly. Five years ago, only humans were using algorithms; now, some algorithms are also using other algorithms. This spillover can have serious consequences in terms of diluting liability, highlighting the importance of this topic.

Declarations

Conflicts of interest: The authors declare that they have no conflict of interest.

Bibliography

- Agrawal, A., Gans, J. S., and Goldfarb, A. (2024). Artificial intelligence adoption and system-wide change. Journal of Economics & Management Strategy, 33(2):327–337.
- Argenton, C., Potters, J., and Yang, Y. (2023). Receiving credit: On delegation and responsibility. European Economic Review, page 104522.
- Awad, E., Levine, S., Kleiman-Weiner, M., Dsouza, S., Tenenbaum, J. B., Shariff, A., Bonnefon, J.-F., and Rahwan, I. (2020). Drivers are blamed more than their automated cars when both make mistakes. Nature human behaviour, 4(2):134–143.
- Bartling, B. and Fischbacher, U. (2012). Shifting the blame: On delegation and responsibility. The Review of Economic Studies, 79(1):67–87.
- Bartosiak, M. L. and Modlinski, A. (2022). Fired by an algorithm? exploration of conformism with biased intelligent decision support systems in the context of workplace discipline. Career Development International, 27(6/7):601–615.
- Bénabou, R. and Tirole, J. (2005). Self-confidence and personal motivation. Psychology, Rationality and Economic Behaviour: Challenging Standard Assumptions, pages 19–57.
- Bigman, Y. E., Wilson, D., Arnestad, M. N., Waytz, A., and Gray, K. (2023). Algorithmic discrimination causes less moral outrage than human discrimination. Journal of Experimental Psychology: General, 152(1):4.
- Bonnefon, J.-F., Shariff, A., and Rahwan, I. (2016). The social dilemma of autonomous vehicles. Science, 352(6293):1573–1576.
- Burton, J. W., Stein, M.-K., and Jensen, T. B. (2020). A systematic review of algorithm aversion in augmented decision making. Journal of behavioral decision making, 33(2):220–239.
- Cao, S., Jiang, W., Yang, B., and Zhang, A. L. (2023). How to talk when a machine is listening: Corporate disclosure in the age of ai. The Review of Financial Studies, 36(9):3603–3642.

- Castelo, N., Bos, M. W., and Lehmann, D. R. (2019). Task-dependent algorithm aversion. Journal of Marketing Research, 56(5):809–825.
- Chen, D. L., Schonger, M., and Wickens, C. (2016). otree—an open-source platform for laboratory, online, and field experiments. Journal of Behavioral and Experimental Finance, 9:88–97.
- Chevrier, M., Corgnet, B., Guerci, E., and Rosaz, J. (2024). Algorithm Credulity: Human and Algorithmic Advice in Prediction Experiments. GREDEG Working Papers 2024-03, Groupe de REcherche en Droit, Economie, Gestion (GREDEG CNRS), Université Côte d’Azur, France.
- Chugunova, M. and Luhan, W. J. (2024). Ruled by robots: Preference for algorithmic decision makers and perceptions of their choices. Public Choice, pages 1–24.
- Chugunova, M. and Sele, D. (2022). We and it: An interdisciplinary review of the experimental evidence on how humans interact with machines. Journal of Behavioral and Experimental Economics, 99:101897.
- Coeckelbergh, M. (2020). AI ethics. Mit Press.
- Coffman, L. C. (2011). Intermediation reduces punishment (and reward). American Economic Journal: Microeconomics, 3(4):77–106.
- Corgnet, B. (2023). An experimental test of algorithmic dismissals.
- Cowgill, B. and Tucker, C. E. (2019). Economics, fairness and algorithmic bias. preparation for: Journal of Economic Perspectives.
- Dana, J., Weber, R. A., and Kuang, J. X. (2007). Exploiting moral wiggle room: experiments demonstrating an illusory preference for fairness. Economic Theory, 33:67–80.
- Dargnies, M.-P., Hakimov, R., and Kübler, D. (2024). Aversion to hiring algorithms: Transparency, gender profiling, and self-confidence. Management Science.
- Daschner, S. and Obermaier, R. (2022). Algorithm aversion? on the influence of advice accuracy on trust in algorithmic advice. Journal of Decision Systems, 31(sup1):77–97.
- Dawes, R. M. (2008). The robust beauty of improper linear models in decision making. In Rationality and Social Responsibility, pages 321–344. Psychology Press.

- Dietvorst, B. J., Simmons, J. P., and Massey, C. (2015). Algorithm aversion: people erroneously avoid algorithms after seeing them err. Journal of experimental psychology: General, 144(1):114.
- Dinet, J. and Vivian, R. (2014). Exploratory investigation of attitudes towards assistive robots for future users. Le travail humain, 77(2):105–125.
- Dohmen, T., Falk, A., Huffman, D., Sunde, U., Schupp, J., and Wagner, G. G. (2011). Individual risk attitudes: Measurement, determinants, and behavioral consequences. Journal of the european economic association, 9(3):522–550.
- Donnellan, M. B. and Robins, R. W. (2010). Resilient, overcontrolled, and undercontrolled personality types: Issues and controversies. Social and Personality Psychology Compass, 4(11):1070–1083.
- Fehr, E. and Schmidt, K. M. (2006). The economics of fairness, reciprocity and altruism—experimental evidence and new theories. Handbook of the economics of giving, altruism and reciprocity, 1:615–691.
- Feier, T., Gogoll, J., and Uhl, M. (2022). Hiding behind machines: Artificial agents may help to evade punishment. Science and Engineering Ethics, 28(2):19.
- Feiler, L. (2014). Testing models of information avoidance with binary choice dictator games. Journal of Economic Psychology, 45:253–267.
- Fershtman, C. and Gneezy, U. (2001). Strategic delegation: An experiment. RAND Journal of Economics, pages 352–368.
- Fildes, R. and Goodwin, P. (2007). Against your better judgment? how organizations can improve their use of management judgment in forecasting. Interfaces, 37(6):570–576.
- Fischer, J. M. (1999). Recent work on moral responsibility. Ethics, 110(1):93–139.
- Frederick, S. (2005). Cognitive reflection and decision making. Journal of Economic perspectives, 19(4):25–42.
- Garcia, T., Massoni, S., and Villeval, M. C. (2020). Ambiguity and excuse-driven behavior in charitable giving. European Economic Review, 124:103412.

- Gawn, G. and Innes, R. (2021). Machiavelli preferences without blame: Delegating selfish vs. generous decisions in dictator games. Journal of Behavioral and Experimental Economics, 90:101615.
- Gazit, L., Arazy, O., and Hertz, U. (2023). Choosing between human and algorithmic advisors: The role of responsibility sharing. Computers in Human Behavior: Artificial Humans, 1(2):100009.
- Gerken and McMahon (21 September 2023). Game of thrones author sues chatgpt owner openai. BBC, pages 1–14.
- Goolsbee, A. (2018). Public Policy in an AI Economy, pages 309–316. University of Chicago Press.
- Greiner, B. (2015). Subject pool recruitment procedures: organizing experiments with orsee. Journal of the Economic Science Association, 1(1):114–125.
- Grossman, Z. and Van Der Weele, J. J. (2017). Self-image and willful ignorance in social decisions. Journal of the European Economic Association, 15(1):173–217.
- Hamman, J. R., Loewenstein, G., and Weber, R. A. (2010). Self-interest through delegation: An additional rationale for the principal-agent relationship. American Economic Review, 100(4):1826–1846.
- Hertz, N. and Wiese, E. (2019). Good advice is beyond all price, but what if it comes from a machine? Journal of Experimental Psychology: Applied, 25(3):386.
- Hou, Y. T.-Y. and Jung, M. F. (2021). Who is the expert? reconciling algorithm aversion and algorithm appreciation in ai-supported decision making. Proceedings of the ACM on Human-Computer Interaction, 5(CSCW2):1–25.
- Jung, M. and Seiter, M. (2021). Towards a better understanding on mitigating algorithm aversion in forecasting: An experimental study. Journal of Management Control, 32(4):495–516.
- Jussupow, E., Benbasat, I., and Heinzl, A. (2020). Why are we averse towards algorithms? a comprehensive literature review on algorithm aversion.
- Kassam, A. and Marino, P. (2022). Algorithmic racial discrimination: A social impact approach. Feminist Philosophy Quarterly, 8(3/4).

- Kirchkamp, O. and Strobel, C. (2019). Sharing responsibility with a machine. Journal of Behavioral and Experimental Economics, 80:25–33.
- Kleinberg, J., Ludwig, J., Mullainathan, S., and Sunstein, C. R. (2018). Discrimination in the age of algorithms. Journal of Legal Analysis, 10:113–174.
- Klockmann, V., von Schenk, A., and Villeval, M. C. (2022). Artificial intelligence, ethics, and intergenerational responsibility. Journal of Economic Behavior & Organization, 203:284–317.
- Langer, M. and Landers, R. N. (2021). The future of artificial intelligence at work: A review on effects of decision automation and augmentation on workers targeted by algorithms and third-party observers. Computers in Human Behavior, 123:106878.
- Larson, T. and Capra, C. M. (2009). Exploiting moral wiggle room: Illusory preference for fairness? a comment. Judgment and decision Making, 4(6):467–474.
- Lee, M. K. (2018). Understanding perception of algorithmic decisions: Fairness, trust, and emotion in response to algorithmic management. Big Data & Society, 5(1):2053951718756684.
- Lehmann, C. A., Haubitz, C. B., Fügener, A., and Thonemann, U. W. (2022). The risk of algorithm transparency: How algorithm complexity drives the effects on the use of advice. Production and Operations Management, 31(9):3419–3434.
- Leib, M., Köbis, N., Rilke, R. M., Hagens, M., and Irlenbusch, B. (2024). Corrupted by algorithms? how ai-generated and human-written advice shape (dis) honesty. The Economic Journal, 134(658):766–784.
- Lessmann, S., Baesens, B., Seow, H.-V., and Thomas, L. (2015). Benchmarking state-of-the-art classification algorithms for credit scoring: An update of research. European Journal of Operational Research, (doi:10.1016/j.ejor.2015.05.030).
- Lipinski, T. A., Buchanan, E. A., and Britz, J. J. (2002). Sticks and stones and words that harm: Liability vs. responsibility, section 230 and defamatory speech in cyberspace. Ethics and Information Technology, 4:143–158.

- Logg, J. M., Minson, J. A., and Moore, D. A. (2019). Algorithm appreciation: People prefer algorithmic to human judgment. Organizational Behavior and Human Decision Processes, 151:90–103.
- Mahmud, H., Islam, A. N., Ahmed, S. I., and Smolander, K. (2022). What influences algorithmic decision-making? a systematic literature review on algorithm aversion. Technological Forecasting and Social Change, 175:121390.
- Manlove, D. (2013). Algorithmics of matching under preferences, volume 2. World Scientific.
- Martin, K. (2019). Ethical implications and accountability of algorithms. Journal of business ethics, 160:835–850.
- Matthias, A. (2004). The responsibility gap: Ascribing responsibility for the actions of learning automata. Ethics and information technology, 6:175–183.
- May, L. (1992). Sharing responsibility. University of Chicago Press.
- Meehl, P. E. (1954). Clinical versus statistical prediction: A theoretical analysis and a review of the evidence.
- Meijer, A. and Wessels, M. (2019). Predictive policing: Review of benefits and drawbacks. International Journal of Public Administration.
- Nomura, T., Kanda, T., and Suzuki, T. (2006). Experimental investigation into influence of negative attitudes toward robots on human–robot interaction. Ai & Society, 20:138–150.
- Oexl, R. and Grossman, Z. J. (2013). Shifting the blame to a powerless intermediary. Experimental Economics, 16:306–312.
- Otero, I. and Alonso, P. (2023). Cognitive reflection test: The effects of the items sequence on scores and response time. PLOS ONE, 18(1):1–15.
- Papachristou, M., Banerjee, S., and Kleinberg, J. (2024). Optimal resource allocation for remediating networked contagions. Available at SSRN 4880536.

- Primi, C., Morsanyi, K., Chiesi, F., Donati, M. A., and Hamilton, J. (2016). The development and testing of a new version of the cognitive reflection test applying item response theory (irt). Journal of Behavioral Decision Making, 29(5):453–469.
- Rahwan, I., Cebrian, M., Obradovich, N., Bongard, J., Bonnefon, J.-F., Breazeal, C., Crandall, J. W., Christakis, N. A., Couzin, I. D., Jackson, M. O., et al. (2019). Machine behaviour. Nature, 568(7753):477–486.
- Rao, J. M. and Reiley, D. H. (2012). The economics of spam. Journal of Economic Perspectives, 26(3):87–110.
- Sanders, N. R. and Manrodt, K. B. (2003). The efficacy of using judgmental versus quantitative forecasting methods in practice. Omega, 31(6):511–522.
- Sengul, M., Gimeno, J., and Dial, J. (2012). Strategic delegation: A review, theoretical integration, and research agenda. Journal of Management, 38(1):375–414.
- Shariff, A., Bonnefon, J.-F., and Rahwan, I. (2017). Psychological roadblocks to the adoption of self-driving vehicles. Nature Human Behaviour, 1(10):694–696.
- Shrestha, Y. R., Ben-Menahem, S. M., and Von Krogh, G. (2019). Organizational decision-making structures in the age of artificial intelligence. California management review, 61(4):66–83.
- Stieger, S. and Reips, U.-D. (2016). A limitation of the cognitive reflection test: Familiarity. PeerJ, 4:e2395.
- Sutan, A. and Vranceanu, R. (2016). Lying about delegation. Journal of Economic Behavior & Organization, 121:29–40.
- Toplak, M. E., West, R. F., and Stanovich, K. E. (2014). Assessing miserly information processing: An expansion of the cognitive reflection test. Thinking & reasoning, 20(2):147–168.
- Van Esch, P., Black, J. S., and Ferolie, J. (2019). Marketing ai recruitment: The next phase in job application and selection. Computers in Human Behavior, 90:215–222.
- von Schenk, A., Klockmann, V., Bonnefon, J.-F., Rahwan, I., and Köbis, N. (2022). Lie detection algorithms attract few users but vastly increase accusation rates. arXiv preprint arXiv:2212.04277.

- Wolff, W., Bieleke, M., Englert, C., Bertrams, A., Schüller, J., and Martarelli, C. S. (2022). A single item measure of self-control–validation and location in a nomological network of self-control, boredom, and if-then planning. Social Psychological Bulletin, 17:1–22.
- You, S., Yang, C. L., and Li, X. (2022). Algorithmic versus human advice: does presenting prediction performance matter for algorithm appreciation? Journal of Management Information Systems, 39(2):336–365.
- Zhang, Y., Liao, Q. V., and Bellamy, R. K. (2020). Effect of confidence and explanation on accuracy and trust calibration in ai-assisted decision making. In Proceedings of the 2020 conference on fairness, accountability, and transparency, pages 295–305.
- Zizzo, D. J. (2010). Experimenter demand effects in economic experiments. Experimental Economics, 13:75–98.

A Instructions

Instructions translated from French.

Welcome instructions (for every experiment conducted at LEEN):

Welcome to the Experimental Economics Laboratory of Nice (the LEEN). By agreeing to participate in this experiment, you express your full agreement with the Laboratory’s regulations, available on the website or upon request. You will participate in an experiment where your decisions will be anonymous and will partly determine your final payment. Therefore, we invite you to read the following instructions carefully. In addition to earnings collected in the experiment and regardless of your decisions, a fixed amount of 5 euros will be given to you to cover your travel expenses. A variable amount will be added based on your decisions during the experiment. Your earnings will be paid individually and confidentially at the end of the experiment. In order not to alter the results of the experiment, we ask that you do not communicate with other participants. We also ask that you kindly turn off your mobile phones and refrain from using them throughout the duration of the experiment. If these rules are breached, the experiment

will be interrupted, and earnings will be forfeited. If you encounter a technical problem, please simply raise your hand, and wait for the experimenter to come to you. Everybody in this room has access to the same instructions and will participate in the same experiment.

Experiment starting :

The decisions you make during this experimental session will be anonymous and confidential. Our accounting sheet, which certifies your payment, is the only form where your name and payment will be recorded (example given verbally). You will never be asked to reveal your identity to anyone during the experiment. To maintain the confidentiality of decisions, please do not disclose your choices to other participants.

In this experiment, you will be anonymously grouped with three other participants. Your earnings will be calculated based on your decision and the decisions made by the other participants.

At the end of the experiment, you will receive a variable amount in addition to the €5 participation fee. You will not know the identities of the other participants, and they will not know yours. Each group will consist of four types of participants: participant A, participant B, participant C, and participant D.

Decisions of participants A and B:

Participant A starts. She has the option to choose the distribution of €20 among the four participants. Participant A chooses between three options:

- Distribution 1: Participant A and participant B each receive €9, participants C and D receive €1 each (i.e., $A = €9$, $B = €9$, $C = €1$, $D = €1$).
- Distribution 2: Each participant receives €5 (i.e., $A = €5$, $B = €5$, $C = €5$, $D = €5$).
- Delegation: Participant A can choose to let participant B decide.

[RA/AI Programmer]Participant A can choose to let an algorithm decide.

Regardless of participant A's choice, participant B chooses between the first two distributions:

- Distribution 1: Participant A and participant B each receive €9, participants C and D receive €1 each (i.e., $A = €9$, $B = €9$, $C = €1$, $D = €1$).
- Distribution 2: Each participant receives €5 (i.e., $A = €5$, $B = €5$, $C = €5$, $D = €5$).

If participant A opts for distribution 1 or 2, participant B's decision will not be considered. If A lets participant B decide, the distribution chosen by B is the one that will be applied.

[RA/AI Programmer] If A delegates, it is the algorithm that will decide. This decision will be based on B choice.

[RA Programmer] In 100% of cases, the algorithm will apply participant B's decision. In 0% of cases, the algorithm makes the decision itself.

[AI Programmer] In 50% of cases, the algorithm will apply participant B's decision. In 50% of cases, the algorithm makes the decision itself.

The algorithm bases its decision on data collected from subjects who have taken part in an experiment similar to this one.

The algorithm uses this data to make its decision and choose between option 1 and option 2. The algorithm simply reproduces the decisions made by individuals.

For example, if the individuals in the database have chosen distribution 1 90% of the time, then the algorithm has a 90% chance of choosing distribution 1 and a 10% chance of choosing decision 2.

Decisions of participants C and D:

Participants C and D have the option to reduce the earnings of the other participants by agreeing to pay €1. If they agree, they can reduce the earnings of one participant by up to €7 or distribute it among all other participants. It is not mandatory to distribute the entire €7. The participant can decide to distribute only part of the €7. It is important to note that participants C and D cannot subtract more than what a participant has earned from the chosen distribution (each participant will receive at least the €5

participation fee).

In total, 4 scenarios are possible:

1. Participant A chooses distribution 1.
2. Participant A chooses distribution 2.
3. Participant A delegates to B, and B chooses distribution 1.

[RA/AI Programmer] Participant A delegates to the algorithm, algorithm chooses distribution 1

4. Participant A delegates to B, and B chooses distribution 2.

[RA/AI Programmer] Participant A delegates to the algorithm, algorithm chooses distribution 2

Participants C and D must make their decision based on each of these 4 scenarios. Depending on the decisions made by participants A and B [RA/AI Programmer]the algorithm, one of these scenarios will be selected. Consequently, the same scenario will be selected for C and D. At the end of the experiment, either participant C or D will be randomly chosen. Only the decision of the participant (C or D) who was selected will be applied.

For example:

Imagine participant A chooses to delegate to participant B. And participant B decides to choose distribution 1 ($A = \text{€}9$, $B = \text{€}9$, $C = \text{€}1$, and $D = \text{€}1$). Meanwhile, participants C and D make their choices based on the 4 different scenarios. They are asked to indicate if they want to reduce the payment of the other participants, in the case where:

-Participant A chose distribution 1

-Participant A chose distribution 2

-Participant A delegates, and B chose distribution 1

[RA/AI Programmer]Participant A delegates to the algorithm, algorithm chooses distribution 1

-Participant A delegates, and B chose distribution 2

[RA/AI Programmer]Participant A delegates to the algorithm, algorithm chooses distribution 2

The decisions made by participant A and B [RA/AI Programmer] the algorithm correspond to a scenario, here scenario 3. For scenario 3, participants C and D chose: Participant C decides to pay €1 to reduce the earnings of the other participants. Their decision is to reduce participant A's earnings by €2, participant B's earnings by €4, and participant D's earnings by €0. Participant D decides to pay €1 to reduce the earnings of the other participants. Their decision is to reduce participant A's earnings by €7, participant B's and C's earnings by €0.

Let's assume participant C was chosen by random draw. As a result, participant A earns €7 ($€9 - €2$), participant B earns €5 ($€9 - €4$), participant D earns €1, and participant C earns €0 ($€1 - €1$ cost). Let's assume participant D was chosen by random draw.

As a result, participant participant A earns €2 ($€9 - €7$), participant B earns €9 ($€9 - €0$), participant D earns €0 ($€1 - €1$ cost), and participant C earns €1.

Note that the participant whose decision was not drawn will not have to bear the €1 deduction.

Continuation of the Experiment:

Once you have understood the instructions, you will be asked to answer some comprehension questions. Failure to pass this questionnaire will result in exclusion from the session. Once completed, a screen displaying "please wait" will appear, indicating that you must wait for the other participants to make their choices. Once finished, your role (A, B, C, or D) will be assigned to you, and you will be asked to make your choice. After completing the last two questionnaires, you will be informed of your earnings. You will then be asked to wait for the experimenter to come and ask you to leave to proceed with the payment.

Please remain in your seat while waiting.

B Post-experimental questionnaire

B.1 Modify Cognitive Reflection Test 6 items ([Primi et al., 2016](#); [Toplak et al., 2014](#))

Subjects had five minutes to answer the following questions:

- There's an area of water lilies in a lake. Every day, the area doubles in size. It takes 100 days for the area to cover the whole lake. How long would it take for the area to cover a quarter of the lake? (Example: 2 days, 10 days, 120 days, ...)
- If three elves can pack 6 toys in an hour, how many elves are needed to pack 12 toys in 2 hours? (Expected form of answer: 2 elves, 10 elves, 120 elves, ...)
- A coffee and an apple cost a total of €4.50. The coffee costs €4.00 more than the apple. How much does the apple cost? (Example: 0.50; 1.00; 3.00; ...)
- Jerry received both the 20th highest mark and the 20th lowest mark in the class. How many pupils are there in the class? (Expected form of answer: 2 pupils, 10 pupils, 120 pupils, ...)
- In an athletics team, tall members are three times more likely to win a medal than short members. This year, the team has won 80 medals so far. How many of these medals have been won by short athletes? (Expected form of answer: 2 medals, 10 medals, 120 medals, ...)

B.2 Negative Attitude Toward Robot ([Nomura et al., 2006](#))

The following questions measure the subjects' level of sociability with robots. Subjects could choose from 5 levels of response from "Strongly disagree" to "Strongly agree"

- I would feel uncomfortable if I were given an activity in which I had to use a robot.
- The word "robot" means nothing to me.
- I'd feel nervous if I had to use a robot in front of other people.
- I hate the idea of robots making judgements about things.
- I'd feel nervous if I just had to stand up to a robot.
- I'd feel very uncomfortable talking to a robot.

B.3 Risk, ambiguity and control preferences (Dohmen et al., 2011; Wolff et al., 2022)

Licker scale from 0 to 10 voir Table 5

- Would you say that you tend to take risks in your life in general?
(0 means “I never take risks” and 10 means “I take risks all the time”).
- When you have no information about a situation, would you say that you tend to be extremely cautious or extremely reckless in your life in general?
(0 means “I am extremely cautious” and 10 means “I am extremely reckless”).
- When you have no information about a situation, do you consider yourself extremely pessimistic or extremely optimistic about the outcome?
(0 means “I am extremely pessimistic” and 10 means “I am extremely optimistic”).
- On a scale of 1 to 10, how would you rate your attitude to ambiguity?
When faced with a lottery, would you prefer not to have any information about the probabilities (1) or would you prefer to know the exact probability of winning or losing (10)?
- Estimate your level of self-control.
(0 means “I have no self-control” and 10 means “I have very high self-control”).
- Can you let go?
(0 means “I never let go” and 10 means “I let go easily”).

B.4 Socio-demographic

The name of the variable used in Table 4 are in brackets.

- Your age: 18-21; 22-25; 26-29²³ [Age]
- Your gender: *Male*; *Female* [Male]
- Level of education: *Baccalaureate*; *Bachelor*; *Master*; *PhD*.
- Field : “*Biology, Health, Sports*”;
“*Law, Political Science, Economics and Management*” ;
“*Education, Teaching, Training*”;
“*Letters, Languages, Arts and Communication*”;

²³To preserve the anonymity of our participants, the Côte d’Azur University Research Ethics Committee recommends avoiding asking for precise ages and instead using age ranges.

“Sciences, Engineering, Technology, Environment”;

“Human and Social Sciences”

C Supplementary

Table 5: Description of variables used in table 4

Variables	Definition	Mean (std. dev)
RA Programmer	1 if the treatment is RA 0 otherwise	0.33(0.51)
AI programmer	1 if the treatment is AI 0 otherwise	0.33(0.51)
A egalitarian	1 if player was playing the hypothetical scenario where A chose fair 0 otherwise	-
B inegalitarian	1 if player was playing the hypothetical scenario where A delegates and B chose unfair 0 otherwise	-
B egalitarian	1 if player was playing the hypothetical scenario where A delegates and B chose fair 0 otherwise	-
<u>Socio-demographic variables</u>		
Age	Age of subject	18-22
Male	1 if subject is male; 0 otherwise	0.425(0.49)
Risk	Answer from an 11-point Likert scale: 0 for a risk averse person and 10 for a person who is risk lover	6.07(2.15)
Ambiguity	Answer from an 11-point Likert scale: 0 for a person that prefers to play a lottery without knowing probabilities and 10 for perfect information preference.	7.47(2.92)
Control	Answer from an 11-point Likert scale: 0 for a perfect self-control and 10 for absence of self-control	7.14(1.75)
NARS	Score from 0 to 7 resulting for their answer to 7 questions A high score meaning that the subject is averse to robots.	9.10(4.45)
Positive	Answer from an 11-point Likert scale: 0 for a pessimistic person and 10 for a person who is optimistic.	5.9(2.55)
CRT True	Number of correct answers obtained by the participant on the CRT6 Variable ranges between 0 and 6.	2.15(1.69)

D Comparison

Figure 4: Punishment pattern in Delegation & Punishment (D&P) in [Bartling and Fischbacher \(2012\)](#) an punishment pattern for our baseline (Human treatment)

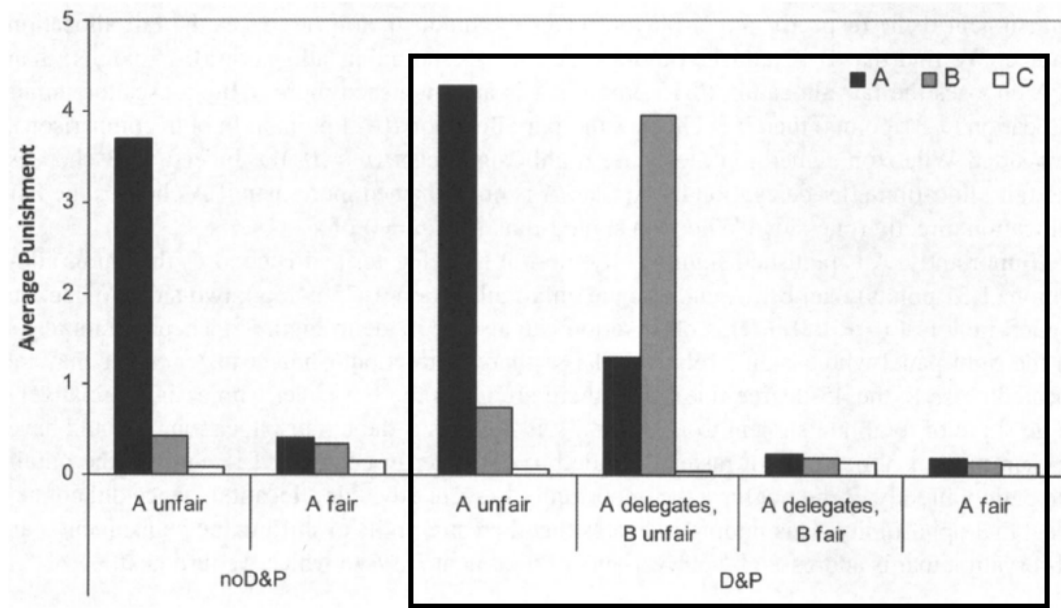
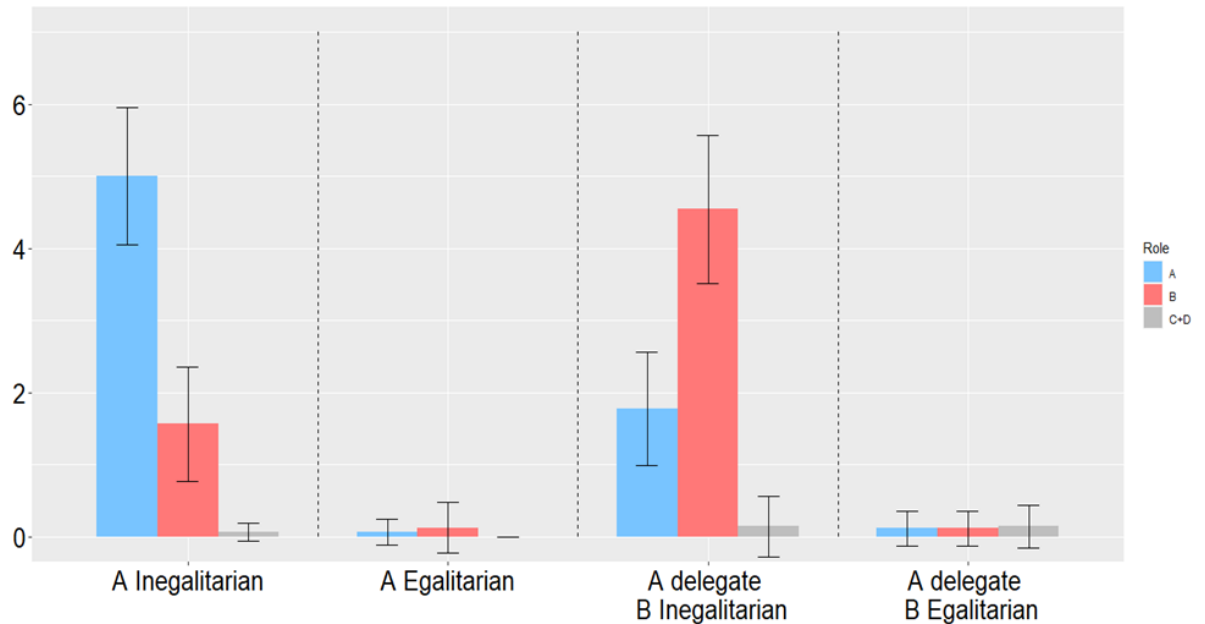


FIGURE 1
Punishment pattern in treatments *noD&P* and *D&P*



In the upper figure squared by the black rectangle, we observe the original treatment. “A Unfair” is represented in our study by “A egalitarian”. “A Unfair” is represented by “A Inegalitarian” and so on. Their result and our result show the same tendency.

Table 6: Probit regression on the decision to delegate

	Principal	Principal
RA Programmer	-0.001 (-0.00)	-0.614 (-1.13)
AI Programmer	0.240 (0.64)	0.192 (0.38)
Age		-1.339* (-2.29)
Male		1.053*** (2.63)
Risk		-0.303*** (-2.78)
Ambiguity		-0.149* (-2.17)
Control		0.0964 (0.93)
CRT True		0.283* (2.21)
Nars		-0.151*** (-3.08)
Control	No	Yes
Constant	-0.967*** (-3.53)	0.762 (0.70)
Observations	90	90

t statistics in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

The delegation decision is a binary variable: 1 if the participant decides to delegate, and 0 if not. Robust standard errors are

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